IS1901:-

Micro-controller based ICT Project

Proposal 1

AbacusGenius 1.0

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ITM - Group 09 (Micro-Explorers)

Group Members

- N.D.B Dissanayake 235034D
- W.S.L.R Perera 235084E
- J.K.P.J Jayawardena 235058E
- S.N Attanagoda 235014R
- S.M.N Rathnayake 235092C

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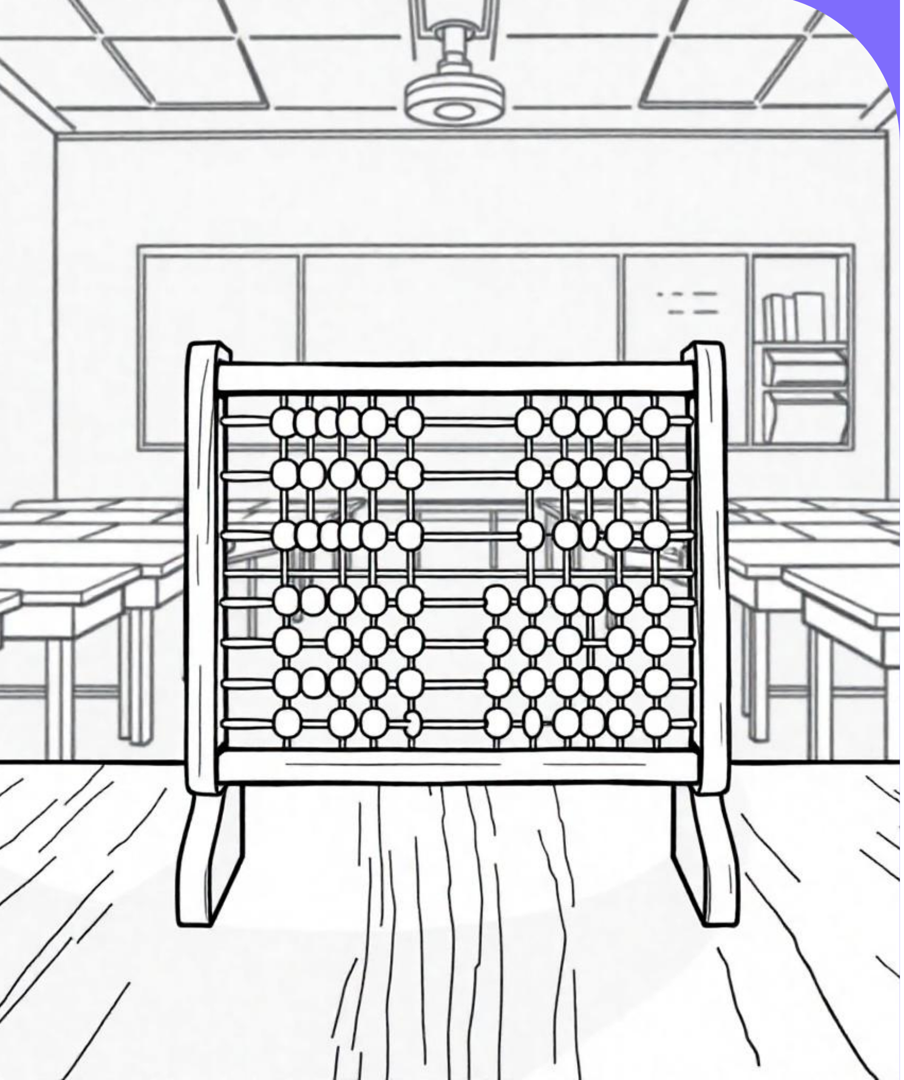
Introduction

Overview of the Electric Abacus

The Electric Abacus enhances the traditional abacus by integrating modern technology such as sensors, displays, and interactive learning modes. This innovative tool combines the tactile experience of manipulating beads with real-time numerical feedback, making arithmetic learning engaging and effective for children.

Significance

The Electric Abacus transforms the learning process, allowing students to grasp mathematical concepts through interactive play while receiving immediate feedback on their calculations.



Problem in Brief

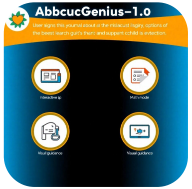
Limitations of Traditional Abacuses: Traditional abacuses serve as static tools primarily designed for basic arithmetic operations. However, they present several limitations:

- **Lack of Interactivity**: They do not offer engaging features that capture children's attention.
- **Absence of Real-Time Feedback**: Users do not receive immediate results or corrections, which is crucial for effective learning.
- Need for a Solution: There is a pressing need for a system that merges physical interaction with technology, providing instant feedback and creating an engaging learning experience for students.

Aim and Features

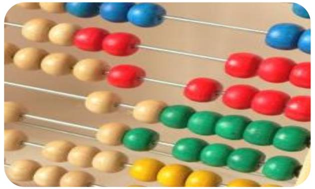
Objectives

The aim of AbacusGenius-1.0 is to create an interactive learning environment that enhances arithmetic skills in children with interactive learning way over the traditional way



Key Features

1. **Interactive Learning**
2. **Different Modes**
3. **Real-Time Calculation**
4. **Visual Guidance**
5. **Learning Game Mode**
6. **Portable Design**.



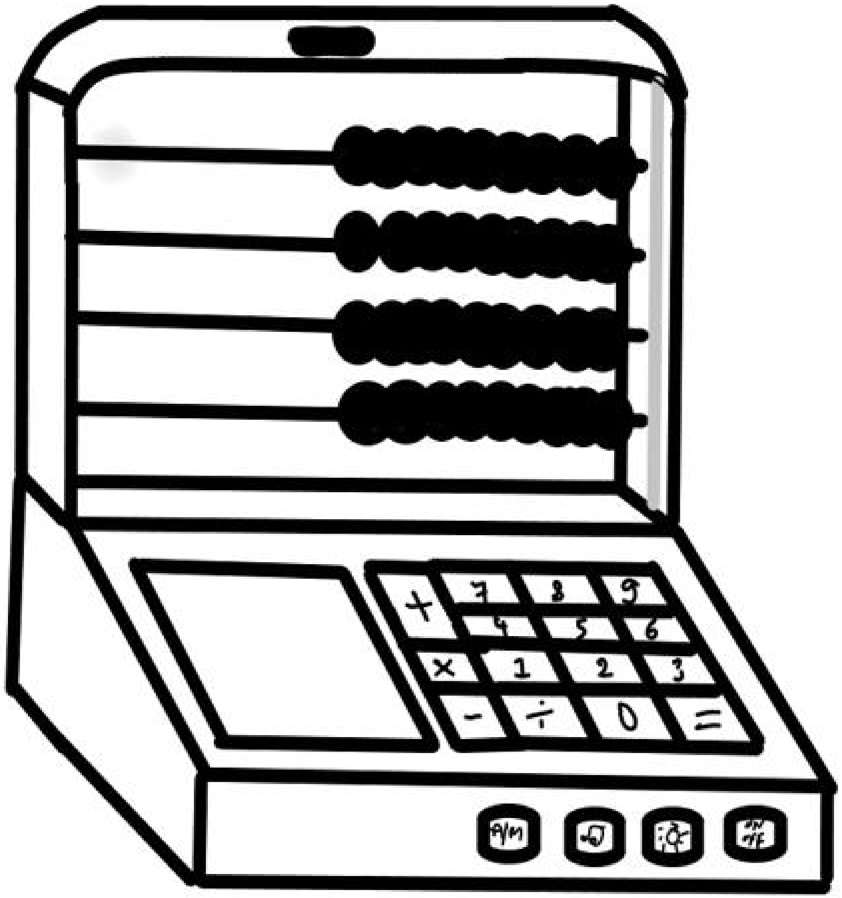
2.Different Modes(further details)

1.User can enter the problem.

2.User can ask for problems.

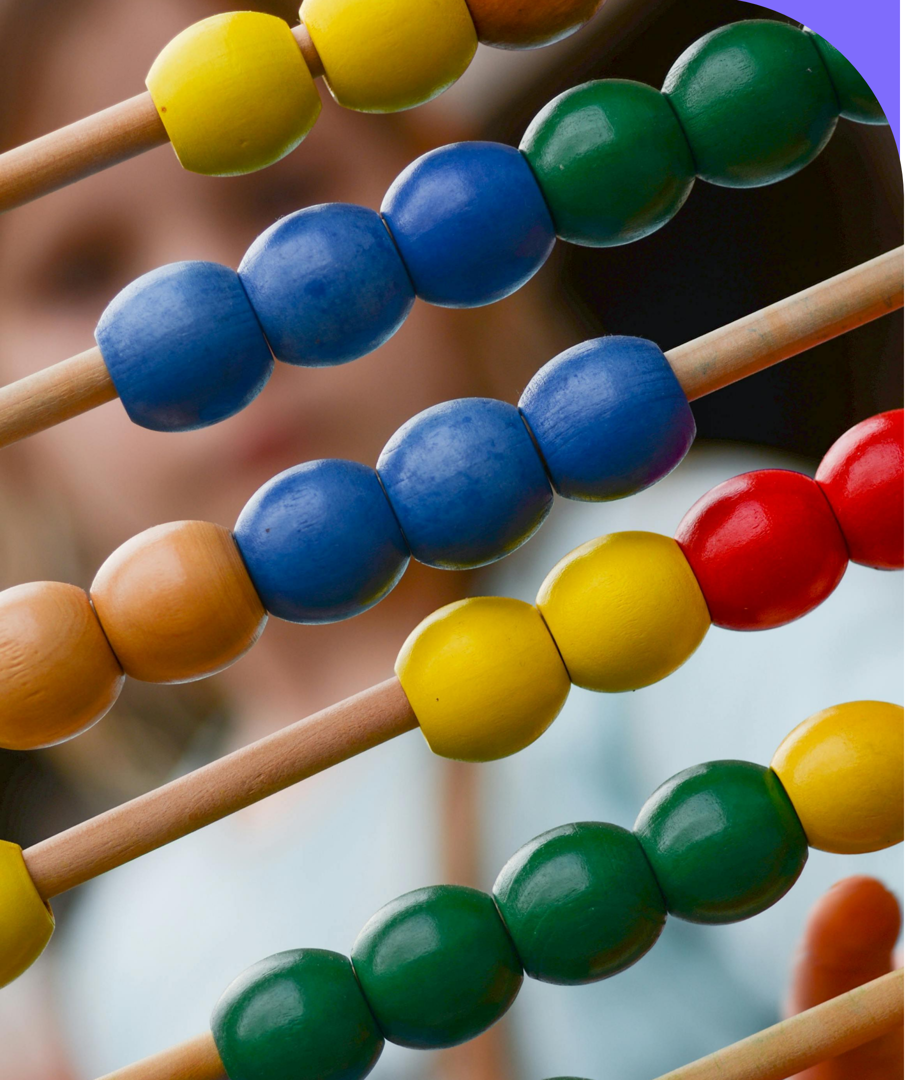
User can choose;

- Whether it is number identification, addition or subtraction.
- Difficulty level



Required Components with Function

- **Abacus Frame**: A sturdy wooden or plastic frame that holds the rods and beads.
- **Beads**: Colourful beads of various sizes that slide along the rods for visual clarity.
- **Sensors**:
- Touch Sensors: Wake the device from sleep mode.
- Light Sensors: Adjust functionality based on ambient light.
- Number Pad: Allows user input.
- PIR Sensor: Detects user presence to manage power.
- **Microcontroller:** An *Arduino Mega NodeMCU WeMos ESP8266* processes sensor inputs and controls logic.



- **Display**: LED or LCD screen shows numeric values and math problems.
- **LED**: Illuminates the abacus for nighttime use.
- **Speakers**: Optional sound feedback for user interactions.
- **Battery**: Rechargeable power source for portability.
- **Encoder**: Counts beads on each row for accurate output.
- **Neodymium Magnets**: Facilitate smooth bead movement and accurate counting.
- **Tiny Iron Balls**: Enable the innovative counting mechanism by completing circuits
- **Connector.**

Neodymium Magnets

Magnets implement to the right and left corners of each row

Model: D82-N52
Dimensions: 1/2"
diameter x 1/8" thick
Grade: N52
Magnetic Field: Strong enough to cover a 19cm area

Integrate neodymium magnets on the right and left corners of the abacus frame. These magnets will prevent beads from getting stuck halfway by ensuring they move smoothly to their correct positions. The magnetic power spreads to a 19cm area only, so it will not interfere with the beads on the opposite (left) corner. This design enhances the accuracy and reliability of the Smart Abacus, providing a seamless counting experience.

Neodymium Magnets (contd....)

Magnets which implement to the inner ring wall of each bead

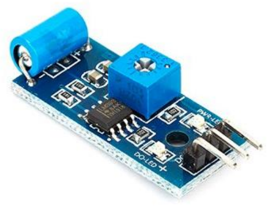
Model: D21-N52
Dimensions: 1/4“
diameter x 1/16" thick
Grade: N52
Magnetic Field: Strong enough to cover a 1.75cm area without exceeding it

Integrate a small neodymium magnet inside the inner ring wall of each bead. This magnet will attract tiny iron balls stored in the abacus rod, completing a circuit when the bead reaches the right corner. This innovative mechanism ensures accurate counting and real-time feedback on a digital screen.

LED Strip

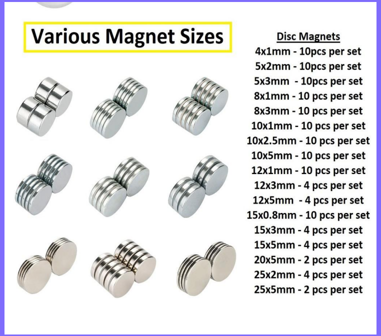
Push Button

Vibration sensor



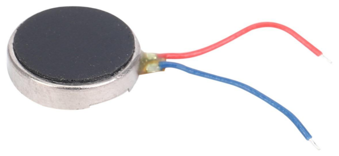
PIR sensor

Neodymium Magnets



Number pad

Vibration motor



Innovative Counting Mechanism for AbacusGenius

Mechanism Overview

The Smart Abacus incorporates a unique counting mechanism with **Holes in the Rod**, **Magnet in Beads**, and **Magnetic Attraction**.

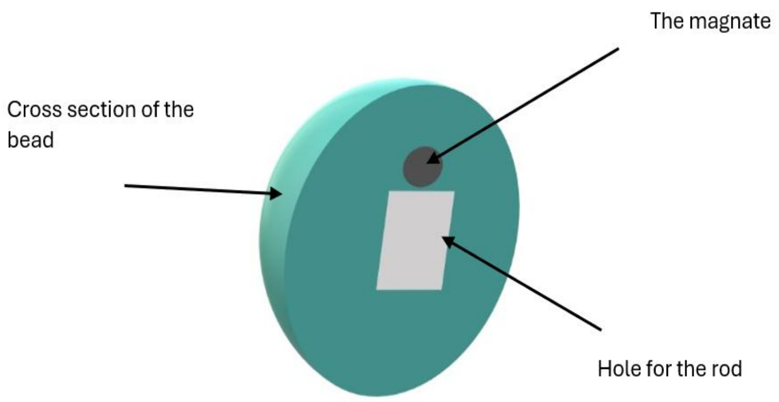
Accurate Counting

Each row has its own encoder to count the beads accurately, and only beads in the correct position generate output.

Benefits

This innovative mechanism not only facilitates accurate counting but also enhances the overall learning experience for children using the Electric Abacus.

The magnet inside the bead



The illustration of the rod of AbacusGenius 1.0

Innovative Counting Mechanism for Smart Abacus

1.**Holes in the Rod**: Store tiny iron balls.

2. **Magnet in Beads**: Attracts iron balls.

3. **Magnetic Attraction**: Bead moves to the right corner, attracts the iron ball.

4. **Enclosed Circuit**: Iron ball completes the circuit, acting as a switch.

5. **Switches for Rows**: Counts beads on the right corner. Encoder totals enclosed circuits for each row (4 encoders total). Outputs count to the digital screen.

6. **Passing Beads**: Iron ball lifts temporarily, drops back down after bead passes, breaking the circuit.

7. **Linear Movement**: Bead and rod cuts prevent rolling, ensuring smooth movement.

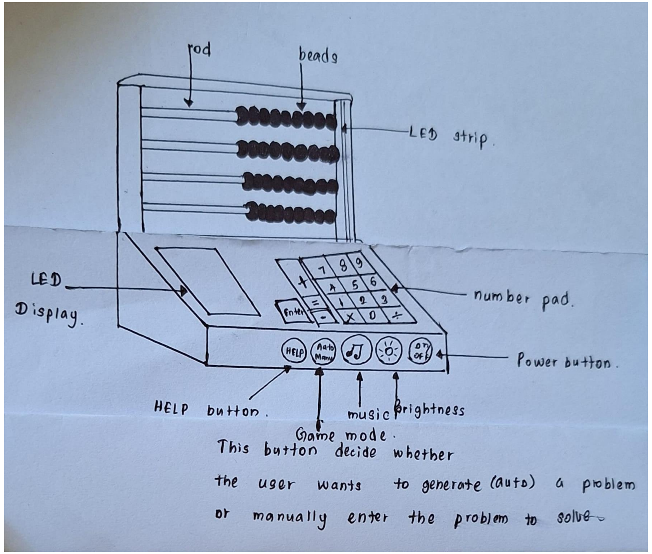
8. **Preventing Magnet Rolling**: Design ensures magnet attracts iron ball to the lid, maintaining the circuit.

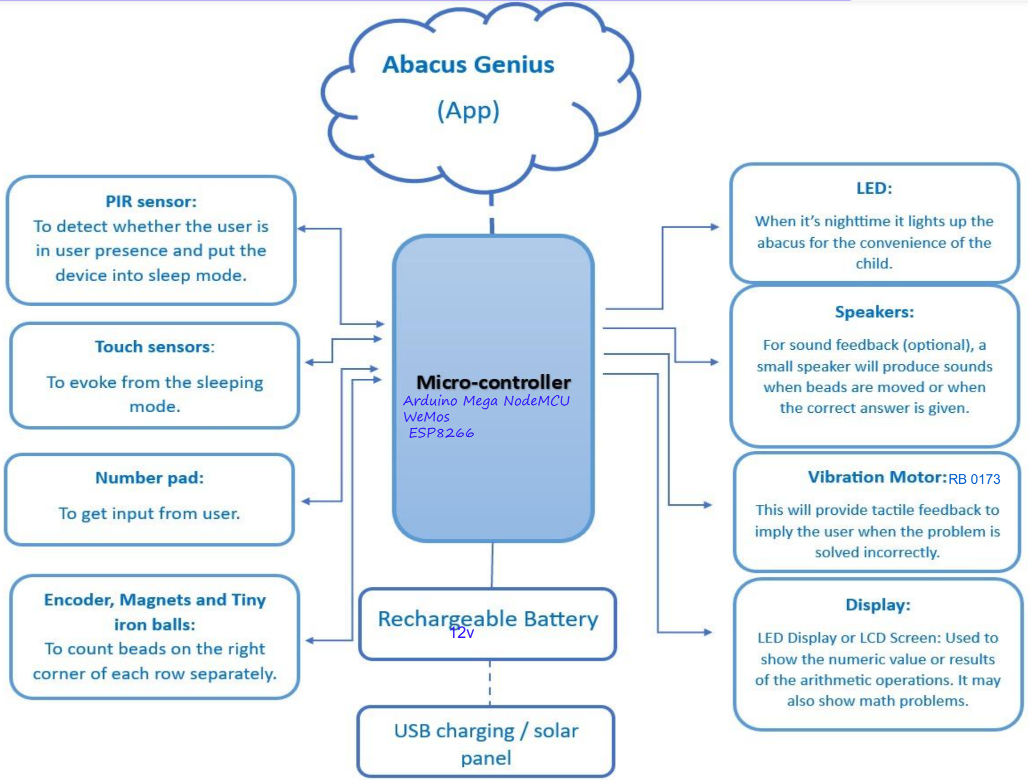
Mobile Application

Features in App:

- Difficulty Levels: Progress through increasingly challenging math problems.
- Performance Records: Keep detailed records of user performance.
- Instant Feedback: Provide real-time feedback on math tests.
- Progress Tracking: Visual representation of progress through graphs and statistics.
- Adaptive Math Tests: Automatically increase difficulty after correct answers.
- Motivation: Show progress to motivate children.

Design & Diagram





Estimated Budget for the Project

3.4 Estimated Budget for the Project

Component	Unit Price	Quantity	Price
Arduino Mega NodeMCU WeMos ESP8266	Rs.4550	1	Rs.4550.00
Encoders	Rs.70.00	4	Rs.280.00
Steel Balls	Rs.50.00	36	Rs.1800.00
Touch Sensor	Rs.320.00	1	Rs.320.00
Light Sensor	Rs.100.00	1	Rs.100.00
Neodymium Magnates	Rs.30.00	36	Rs.1080.00
N52 Disc Magnates	Rs.100.00	8	Rs.800.00
PIR Sensor	Rs.280.00	1	Rs.280.00
LED Display	Rs.400.00	1	Rs.400.00
LED Strips	Rs.350.00	1	Rs.350.00
Connectors	Rs.1500.00	As needed	Rs.1500.00
Buzzer	Rs.80.00	1	Rs.80.00
Vibration Motor	Rs.200.00	1	Rs.200.00
Speaker	Rs.50.00	1	Rs.50.00
Rechargeable Battery	Rs.520.00	1	Rs.520.00
Push Button Switches	Rs.130.00	4	Rs.520.00
Number Pad	Rs.1000.00	1	Rs.1000.00
TOTAL			Rs.14784.00

Conclusion

The Electric Abacus presents an innovative approach to teaching arithmetic by combining traditional principles with modern technology. It offers real-time feedback and various learning modes, making it a valuable educational tool for both classrooms and homes.

By fostering an engaging learning environment, Abacus-Genius can significantly enhance children's arithmetic skills and encourage a love for mathematics.

References

1. Raspberry Pi Pico. (n.d.). Retrieved from [Raspberry Pi](https://www.raspberrypi.org/products/raspberry-pi-pico/)

2. D82- N52 neodymium magnets
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3. D21-N52 neodymium Magnets
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